SS7 and Sigtran Signaling

Course objectives:

The Signaling System No.7, or SS7 in short, provides the foundation for signaling procedures in Telecommunication Networks. Originally developed to manage voice calls in fixed-line telephone networks it rapidly expanded into supporting service add-on's, such as Call Hold, Call Waiting, Number Presentation services and many more.

Mobile Networks had adopted SS7 not only for its versatility and modular design, but also to provide full compatibility with fixed-line network services.

Entering into the era of IP networks required the transport of SS7 signaling messages over IP networks. Signaling Transport (Sigtran) addresses this transition by providing new concepts, functions and adaptation.

This training course will provide participants with a strong foundation in both SS7 and Sigtran.

Starting with signaling principles, we shall take a closer look at the foundation of modern signaling systems and their functional definition with the help of the OSI model.

Participants learn the functions and architecture of each protocol. With the help of the relevant ITU-T technical specification papers, participants apply the knowledge for a basic protocol analysis, using signaling procedure samples collected from live networks.

Duration Instructional methods:

Option 1: On-Site Training: 4 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 4 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 8 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

Telecommunication and Mobile Operators technical staff, support engineers, software developers and anyone interested in signaling procedures and application
It will also benefit to those in the wider technical community who have the need to understand the SS7 / Sigtran protocols, their features and capabilities.

Course Contents:

Introduction to Signaling

- History of Signaling
- Network Reference Architecture for Signaling
- The ISO / OSI Model and its application to Signaling

SS7 Signaling Protocol

- Requirements, Architecture and Functions
- The MTP – Message Transfer Part
- MTP Layer 1 – Functions and Application
- MTP Layer 2 – Functions and Application
- MTP Layer 3 – Functions and Application
- Signaling Message Format in MTP

- SCCP – Signaling Connection Control Part
- Architecture, Functions and Application
- Global Title Routing vs. Direct Address
- SCCP Connectionless Network Services (CLNS)
- SCCP Connection-Oriented Network Services (CONS)
- SCCP Signaling Message Format and practical analysis

- TCAP – Transaction Capabilities Application Part
- Architecture, Function and Application
- Transaction Portion, Dialogue Portion and Component Portion in TCAP
- TCAP Message Format and practical analysis

Network Specific Signaling Protocols

- ISUP (ISDN User Part) and BICC (Bearer-Independent Call Control)
- ISUP and BICC Architecture, Functions and Application
- Differences between ISUP and BICC
- Signaling Message Formats and practical analysis

- MAP (Mobile Application Part) for GSM and UMTS networks
- 3GPP specific signaling functions for Mobile Networks
- Architecture, Functions and Application
- MAP Message Format and practical analysis

- CAMEL (Customized Applications for Mobile network Enhanced Logic)
- CAMEL Protocol Architecture and Functions
- CAMEL Protocol usage and service examples
- CAMEL Functions and Message Formats

- CAMEL procedures practical analysis

SS7 Signaling over IP Networks – Sigtran

- Sigtran Protocol Requirements, Architecture, Implementation
- Sigtran and the OSI Model

- SCTP – Stream Control Transport Protocol
- Architecture, Functions and Application
- Comparing TCP – UDP – SCTP
- SCTP Message Format
- Practical analysis of SCTP basic functions

- Sigtran User Adaptation (UA) Layers
- M2UA/PA, M3UA, SUA and IUA
- Role, Functions and Application
- Mapping of SS7 MTP functions to UA's

- Case Study 1: ISUP signaling over Sigtran (ISUP over M3UA)
- Case Study 2: BICC signaling over Sigtran (BICC over M3UA)

Pre-requisites:

Attending this training assumes a basic understanding of Telecommunication and Mobile Networks.
Knowledge about the respective network architectures, their elements and functions would be an advantage as would be basic understanding of IP-network concepts.

Course Code: SS7SIG